Engineering Mechanics I

Week 3

Yuxia Yuan

Autonomous Aerial Systems

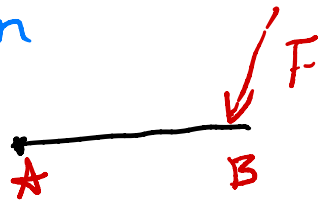
Review: Forces and Moments

Force: effect on a body that changes its internal mechanics of its state of motion

vector: $\vec{F} = m \cdot \vec{a}$ unit: $N = kg \cdot \frac{m}{s^2}$

Moment: Rotational effect of a force acting on a point of a body with respect to a reference point (A) (B)

vector: $\vec{M} = L_{AB} \times \vec{F}$ unit: $N \cdot m$



Equilibrium in statics:

$\sum \vec{F} = 0$

$\sum F_x = 0$

$\sum F_y = 0$

$\sum F_z = 0$

$\sum \vec{M}_A = 0$

$\sum M_{Ax} = 0$

$\sum M_{Ay} = 0$

$\sum M_{Az} = 0$

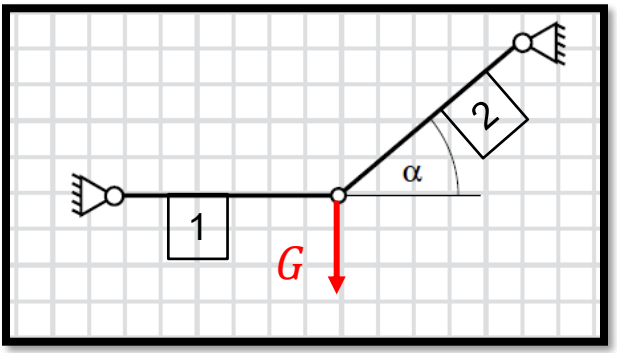
Review: Cutting principle

→ **“Making inner forces visible“**

- Mentally loosening ties by means of free cutting
- Application of all forces and moments acting in the section
- Forces/moments always occur in pairs ("action = reaction")



Exercise 1: Cutting principle 1



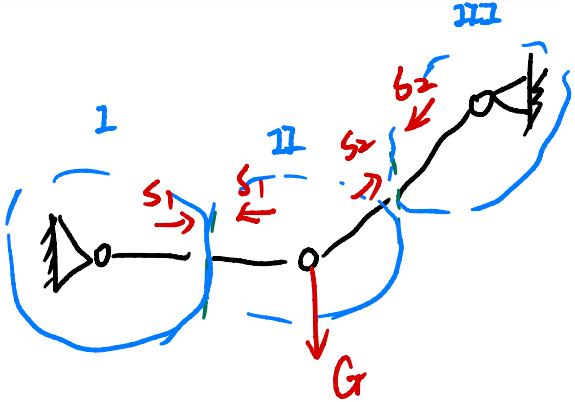
We know:

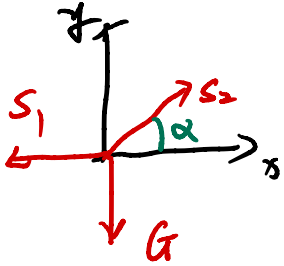
1 and 2 are two ropes, and ropes are massless.

$G = 50N, \alpha = 45^{\circ}$

Calculate:

Rope forces s_1 and s_2





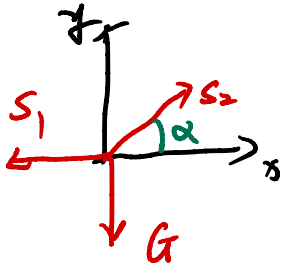
Equilibrium condition/equation

$\sum \vec{F} = 0$

$\sum \vec{F}_x = 0$

$\sum \vec{F}_y = 0$

Exercise 1: Cutting principle 1



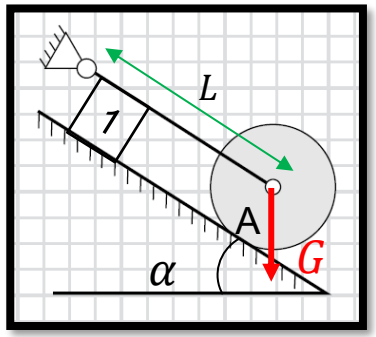
$$\sum \vec{F} = 0$$

$$\sum \vec{F}_x = 0 \quad = \quad -S_1 + S_2 - \cos\alpha = 0$$

$$\sum \vec{F}_y = 0 \quad = \quad -G + S_2 \cdot \sin\alpha = 0$$

$$\Rightarrow \begin{cases} S_1 = 50\ H \\ S_2 = 50\sqrt{2}\ H \end{cases}$$

Exercise 1: Cutting principle 2



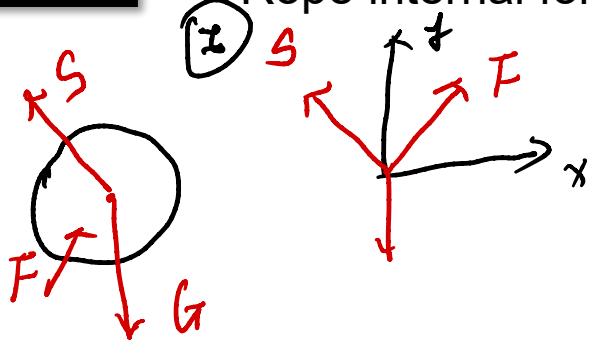
We know:

1 is a rope, and rope is massless. No friction at point A

$L = 2\ \mathrm{m}, G = 40N, \alpha = 30^{\circ}$

Calculate:

Rope internal force s and the contact force at point A



I Equilibrium:

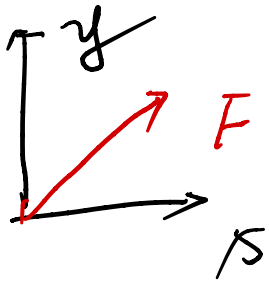
$\sum \vec{F} = 0$

$\sum \vec{F}_x = 0 = G\sin\alpha - S = 0$

$\sum \vec{F}_y = 0 = G\cos\alpha + F = 0$

$\Rightarrow \begin{cases} S = G \cdot \sin\alpha = 20N \\ F = G\cos\alpha = 20\sqrt{3}N \end{cases}$

Review: Support, Support reactions, joints



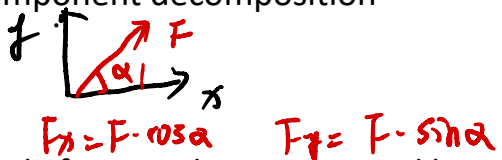
The **directions of the forces** in the supports are **a priori unknown**. Therefore…

- In the free cut, the direction of action is initially assumed to be arbitrary
- The actual direction can be derived from the sign of the calculation results

Procedure

1. Free cut
- Draw support reactions
- Draw joint reactions
- Draw external loads

2. Component decomposition



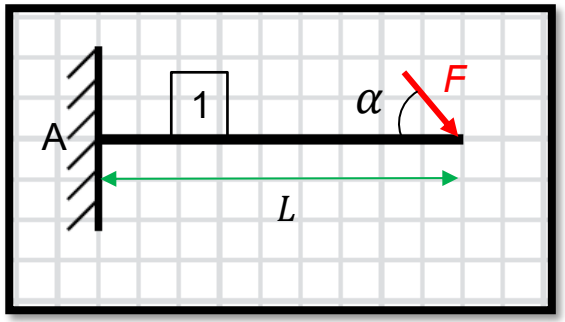
3. Apply force and moment equilibrium

$$\Sigma \vec{F} = 0 \qquad \Sigma \vec{M}_A = 0$$

4. Solve for desired dimensions

5. (Control the entire system)

Exercise 3: Support reactions



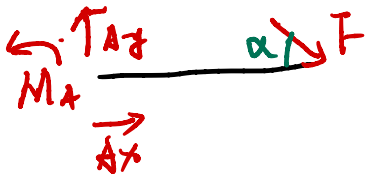
We know:

1 is a beam. External force F, length L, and angle α

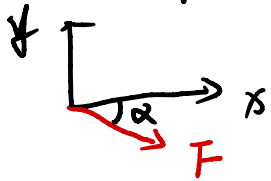
Calculate:

Support reaction on the wall

① Free cut



② decomposition



$F_x = F \cdot \cos\alpha$

$F_y = F \cdot \sin\alpha$

③ Equilibrium

$\Sigma F = 0. \quad \Sigma M_A = 0$

④ calculation.

$\Sigma F_x = A_x + F \cdot \cos\alpha = 0$

$\Sigma F_y = A_y - F \cdot \sin\alpha = 0$

$\Sigma M_A = M_A - L \cdot F \sin\alpha = 0$

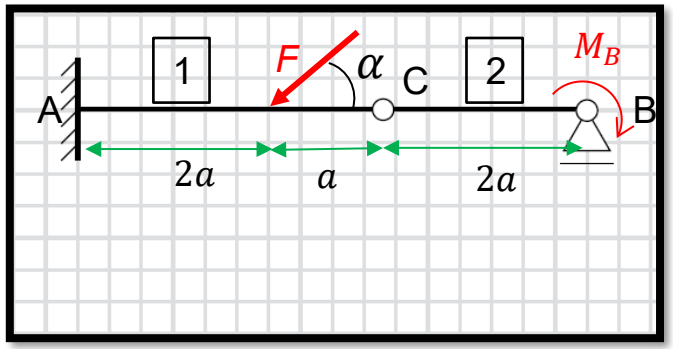
$\Rightarrow$

$A_x = -F\cos\alpha$

$A_y = F\sin\alpha$

$M_A = L \cdot F \sin\alpha$

Exercise 4: Joint reactions

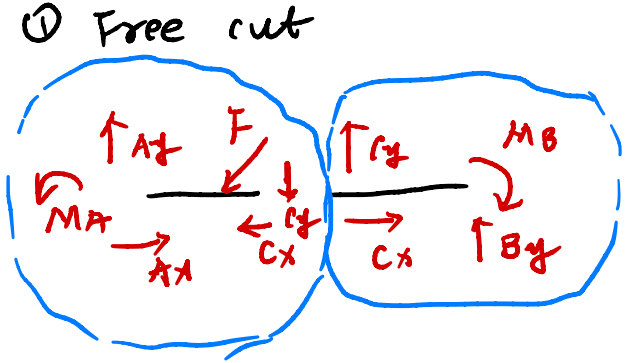


We know:

$F, M_B, \alpha,$ and a

Calculate:

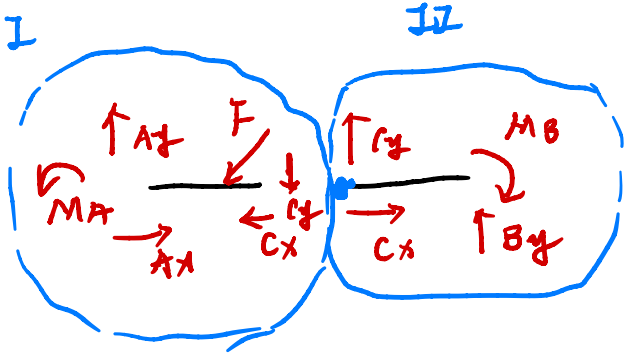
Joint reactions (all internal forces)



Total 2×3 Equilibrium equations

⇒ Equilibrium at point A. B. and C.

Exercise 4: Joint reactions



I
$$\begin{cases} \Sigma F_x = 0 & = A_x - F\cdot\cos\alpha - C_x = 0 \quad (1) \\ \Sigma F_y = 0 & = A_y - F\sin\alpha - C_y = 0 \quad (2) \\ \Sigma M_A = 0 & = M_A - 3a\,C_y - 2a\cdot F\cdot\sin\alpha = 0 \quad (3) \end{cases}$$

II
$$\begin{cases} \Sigma F_x = 0 & = C_x = 0 \quad (4) \\ \Sigma F_y = 0 & = C_y + B_y = 0 \quad (5) \\ \Sigma M_C = 0 & = -M_B + 2a\cdot B_y = 0 \quad (6) \end{cases}$$

$\Rightarrow$

$$A_x = F\cos\alpha$$
$$A_y = F\sin\alpha + \frac{M_B}{2a}$$
$$B_y = \frac{M_B}{2a}$$
$$C_y = \frac{-M_B}{2a}$$
$$M_A = 2aF\cdot\sin\alpha + \frac{3}{2}M_B$$

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